Artificial Intelligence
Lecture 9: An overview of other logics

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SoSe 2025

Reminder: Propositional and predicate logic

Propositional logic gives the tools to describe and reason about single entities:

$$isGreen \rightarrow \neg isRed$$

First-order logic (also called predicate logic) extends this to the description of relations between objects.

$$\forall x [\exists y \ \texttt{isMovie}(x) \ \wedge \ \texttt{directed}(x, \ y) \ ] \ \rightarrow \ \texttt{isDirector}(x)$$

Outline

1. Higher-order logic
2. Modal logic
3. Non-monotonic logics
4. Fuzzy logic
5. Beyond deductive inference

Plan

1 Higher-order logic
- Second-order logic
- Higher-order logic

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Formulae in first-order logic

First-order logic enables to manipulate constant and variables objects and their relations. For instance:

$$\forall x\ (\exists y\ R(x,y) \wedge R(y,x)) \leftrightarrow R(a, f(x))$$

In this sentence:

- x and y are bound variables: they are in the scope of a quantifier;
- a is a constant;
- f is a function of arity 1 (sometimes written $f/1$)
- R is a relation of arity 2 (sometimes written $R/2$)

From first-order to second-order logic

Question: How to express in first-order logic that the function u has an inverse?

Having an inverse means that there exists a function v such that for all $x, u(v(x)) = v(u(x))$. In other words:

$$\exists v \ \forall x \ u(v(x)) = v(u(x))$$

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Question: Is this a valid first-order formula?

No! First-order logic does not allow to quantify over functions nor relations.

Syntax of second-order logic: Symbols

Symbols are the same as for first-order logic, adding some function and relation variables:

- **Variable symbols:** $x, y, z, \ldots$
- **Constant symbols:** a, b, c
- n-ary **function symbols**: $f, g, \ldots$
- n-ary **function variable symbols**: $F, G, \ldots$
- n-ary **predicate symbols:** $P, Q, \ldots$
- n-ary **predicate variable symbols:** $X, Y, \ldots$
- **Logical symbols:**
  - **Logical connectors:** $\neg, \vee, \wedge, \vee, \rightarrow, \leftrightarrow$
  - **Logical quantifiers:** $\forall, \exists$
  - **Equality:** $=$

Syntax of second-order logic: Terms

Terms are similar to terms in first order logic, but using variable functions:

```
term ::=  < variable >
        | < constant >
        | < function >(< term >, < term >, ...)
        | < variable_function >(< term >, < term >, ...)
```

Syntax of second-order logic: Formulas

```
formula ::=  < predicate >(< term >, < term >, ...)
           | < variable_predicate >(< term >, < term >, ...)
           | < term > = < term >
           | ¬ < formula >
           | < formula > < binary_connector > < formula >
           | < quantifier > < variable > < formula >
           | < quantifier > < variable_function > < formula >
           | < quantifier > < variable_predicate > < formula >
```

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Higher-order logic?

Can we do one step further and define a third order logic?

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Why modal logic?

The logics we have encountered so far cannot express ideas like:

- **Necessity vs possibility:** "There is a stench in (1,2)" implies that "*Necessarily* the Wumpus is not in (3,3)" and "*Possibly* the Wumpus is in (1,3)".

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- **Obligation vs permission:** "It is obligatory to choose one action at each time".

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- **Belief:** "I believe that there are two Wumpus in the game".
- **Obligation vs permission:** "It is obligatory to choose one action at each time".
- **Always true vs some day:** "It is always true that there is one Wumpus" and "Some time the agents has felt or will feel a breeze".

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- Syntax
- Semantic
- Axioms and properties

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Syntax

The syntax of modal logic is the same as the syntax of propositional logic, to which are added two unary modal operators: **Box** $\Box$ and **Diamond** $\Diamond$.

$$
\begin{aligned}
< \mathit{Proposition} > \quad &::= \quad < \mathit{AtomicSentences} > \mid < \mathit{CompoundSentences} > \\
< \mathit{AtomicSentences} > \quad &::= \quad < \mathit{PropositionalSymbols} > \\
< \mathit{binaryconnector} > \quad &::= \quad \wedge \mid \vee \mid \rightarrow \mid \leftrightarrow \\
< \mathit{CompoundSentences} > \quad &::= \quad \neg(\mathit{Proposition}) \\
&\quad \mid (< \mathit{proposition} >< \mathit{binaryconnector} >< \mathit{proposition} >) \\
&\quad \mid \Box < \mathit{proposition} > \\
&\quad \mid \Diamond < \mathit{proposition} >
\end{aligned}
$$

What are modal operators?

Depending on the modal logic, $\Box\phi$ will read:

- **Alethic logic:** Necessarily ϕ
- **Temporal logic:** Always ϕ
- **Deontic logic:** ϕ should be true
- **Doxastic logic:** Agent A believes that ϕ
- **Epistemic logic:** Agent A knows that ϕ

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Sets of interpretations

In classical propositional logic, formulae are interpreted over single **interpretations**, i.e. functions mapping each propositional variable to a truth value.

Example: For $\phi = a \vee (a \rightarrow b)$ and $\mathcal{I} = a\bar{b}$, we have $\phi^{\mathcal{I}} = \mathit{True} \vee (\mathit{True} \rightarrow \mathit{False}) = \mathit{True}$.

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- $\Box\phi$ is true with respect to (w, W_w) iff ϕ is true for all accessible worlds $w' \in W_w$.
- $\Diamond\phi$ is true with respect to (w, W_w) iff ϕ is true for some accessible worlds $w' \in W_w$.

An example: Temporal logic

In temporal logic, the context of interest is the time: each world w is given by a time $t \in \mathbb{N}$.

If we take for the set of interpretations that $W = \mathbb{N}$, then the modalities $\Box$ and $\Diamond$ read:

- $\Box\phi$ is true with respect to (w, W) iff ϕ is always true
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If we take $W = \{t \mid t > w\}$, then the modalities $\Box$ and $\Diamond$ read:

- $\Box\phi$ is true with respect to (w, W) iff ϕ will always be true in the future
- $\Diamond\phi$ is true with respect to (w, W) iff ϕ will be true at some point

Kripke models

A **Kripke relational frame** is a pair $\mathcal{F} = \langle W, R \rangle$, where W is a non-empty set of worlds and $R \subseteq W \times W$ is a binary relation called **accessibility relation**.

When $(v, w) \in R$, we say that w is a successor of v, which is written vRw.

A **Kripke model** based on the frame $\mathcal{F} = \langle W, R \rangle$ is a triple $\mathcal{I} = \langle W, R, \pi \rangle$, with π a function mapping worlds to a truth assignment.

Truth in a world

A formula ϕ is true in world w in an interpretation $\mathcal{I} = \langle W, R, \pi \rangle$ under the following conditions:

$\mathcal{I}, w \models a$	iff $\pi_w(a) = \mathit{True}$
$\mathcal{I}, w \models \neg\phi$	iff $\mathcal{I}, w \not\models \phi$
$\mathcal{I}, w \models \phi \wedge \psi$	iff $\mathcal{I}, w \models \phi$ and $\mathcal{I}, w \models \psi$
$\mathcal{I}, w \models \phi \vee \psi$	iff $\mathcal{I}, w \models \phi$ or $\mathcal{I}, w \models \psi$
$\mathcal{I}, w \models \phi \rightarrow \psi$	iff if $\mathcal{I}, w \models \phi$ implies $\mathcal{I}, w \models \psi$
$\mathcal{I}, w \models \phi \leftrightarrow \psi$	iff $\mathcal{I}, w \models \phi$ equivalent to $\mathcal{I}, w \models \psi$
$\mathcal{I}, w \models \Box\phi$	iff $\mathcal{I}, v \models \phi$ for all v such that wRv
$\mathcal{I}, w \models \Diamond\phi$	iff $\mathcal{I}, v \models \phi$ for some v such that wRv

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An important property

Operators $\Box$ and $\Diamond$ are strongly connected by the following equivalences:

$$\Diamond\phi \equiv \neg\Box\neg\phi$$

$$\Box\phi \equiv \neg\Diamond\neg\phi$$

These equivalences are always true. But are there other desired properties?

K axiom

For any frame, we have the following property (called **K axiom**) true:

$$\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$$

Other important axioms

K	$\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$
T	$\Box\phi \to \phi$
4	$\Box\phi \to \Box\Box\phi$
5	$\Diamond\phi \to \Box\Diamond\phi$
B	$\phi \to \Box\Diamond\phi$
D	$\Box\phi \to \Diamond\phi$

Example: Epistemic logic

In epistemic logic, $\Box\phi$ reads "Agent *a* knows ϕ". Conversely, $\Diamond\phi$ reads "Agent *a* thinks ϕ is possible".

K axiom: $\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$ reads "If agent *a* knows that $\phi \to \psi$, then when *a* knows ϕ, then they know ψ".

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Validity of axioms for various modal logics

Logic	$\Box$	$\Diamond$	K	T	4	5	B	D
Epistemic	known	possible	Y	Y	Y	Y	Y	Y
Alethic	necessarily	possibly	Y	Y	Y	Y	Y	Y
Doxastic	believed	possible	Y	N	Y	Y	N	Y
Deontic	obligatory	permitted	Y	N	Y?	Y?	N	Y
Temporal	always in the future	sometimes	Y	Y/N	Y	N	N	N/Y

Axioms and Kripke models

Name	**Axiom**	**Condition on frames**	R **is...**
K	$\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$	-	-
T	$\Box\phi \to \phi$	wRw	reflexive
4	$\Box\phi \to \Box\Box\phi$	If wRv and vRu, then wRu	transitive
5	$\Diamond\phi \to \Box\Diamond\phi$	If wRu and wRv, then uRv	Euclidean
B	$\phi \to \Box\Diamond\phi$	If wRv then vRw	symmetric
D	$\Box\phi \to \Diamond\phi$	for all w, there exists v such that wRv	reflexive

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What is a monotonic logic

Propositional and predicate logics are called **monotonic**, in the sense that:

$$\text{If } KB_1 \subseteq KB_2 \text{ and } KB_1 \models \phi, \text{ then } KB_2 \models \phi$$

Example: If $KB_1 = \{a, a \rightarrow b\}$ and $\phi = b$, we have $KB_1 \models \phi$.

We can extend KB_1 by adding other propositions:

- $KB_2 = KB_1 \cup \{b\}$: In this case, we still have $KB_2 \models \phi$
- $KB_2 = KB_1 \cup \{a \vee b, a \wedge b\}$: In this case, we still have $KB_2 \models \phi$
- $KB_2 = KB_1 \cup \{\neg b\}$: In this case, we KB_2 is not consistent

An example of non-monotonicity: Presumption of innocence

The presumption of innocence is a legal principle stating that a person accused of a crime is considered innocent until proven guilty.

Let's express this principle with an example in the language of first order logic.

$$
\begin{aligned}
KB = \{ \quad & \texttt{accused_of(bob, murder)}, \\
& \texttt{is_crime(murder)}, \\
& \forall x \ \texttt{is_identified_by_witness}(x) \rightarrow \texttt{proven_guilty}(x), \\
& \neg \ \texttt{proven_guilty(bob)}, \\
& \forall x \forall y \ \texttt{accused_of}(x, y) \wedge \texttt{is_crime}(y) \wedge \neg \ \texttt{proven_guilty}(x) \rightarrow \texttt{innocent}(x) \ \}
\end{aligned}
$$

We can check that:

$$KB \models \texttt{innocent(bob)}$$

An example of non-monotonicity: Presumption of innocence

Now, a witness comes and identifies Bob:

`is_identified_by_witness(bob)`

We see that adding this new knowledge has two consequences:

- It is in contradiction with the current state of the knowledge base;
- It overrides what we already knew and makes the statement `innocent(bob)` incorrect wrt the new knowledge base.

Reiter's Default Logic

Default logic is a logic that expresses facts of the form "By default, ϕ is true".

$$\frac{Prerequisite : Justification_1, \ldots, Justication_n}{Conclusion}$$

which interprets as follows:

If the prerequisite is true and that the justifications are consistent with our current beliefs, then we are led to believe that the conclusion is true.

Example of default logic

Example: "A football game will take place unless it is snowing" can be expressed:

$$\frac{FootballGame(X) : \neg IsSnowing(D)}{TakesPlace(X, D)}$$

Question: Why don't we just write $FootballGame(X) \wedge IsSnowing(D) \rightarrow TakesPlace(X, D)$?

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Question: Why don't we just write $FootballGame(X) \wedge IsSnowing(D) \rightarrow TakesPlace(X, D)$?

With this writing, it is necessary to know whether it will snow to actually say anything about the fact that the game will take place. In practice, it is assumed that it will take place by default, unless we get some evidence that it won't.

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An example

Question: What is the truth value of "Bob is a tall man"...

... when Bob is 140cm high? *False*

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Question: What is the truth value of "Bob is a tall man"...

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... when Bob is 195cm high? *True*

An example

Question: What is the truth value of "Bob is a tall man"...

... when Bob is 140cm high? *False*

... when Bob is 195cm high? *True*

... when Bob is 175cm high? *More difficult...*

The set of tall men is not rigidly defined. It is difficult to attribute a binary value indicating whether a man is tall or not.

Fuzzy sets

A **definite set** A can be expressed in the form of its characteristic function:

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

In particular, for all x, we have $\chi_A(x) \in \{0, 1\}$.

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A **fuzzy set** is defined by a **membership function** μ_A such that $\mu_A(x) \in [0, 1]$, with some specific x^* such that $0 < \mu_A(x^*) < 1$.

Fuzzy logic

Propositional logic:

- Interpretations $\mathcal{I}$ attribute a truth value (True or False) to each propositional variable.
- Truth value of compound sentences are defined as functions of the truth values of their parts.

Fuzzy logic follows the same principle, but the truth value of a variable is a real number between 0 and 1.

For instance, $a^{\mathcal{I}} = 0.8$ and $b^{\mathcal{I}} = 0.3$.

Fuzzy connectives

Now that the truth values of propositional variables are chosen, how to compute the truth value of connectives?

For instance, if $a^{\mathcal{I}} = 0.8$ and $b^{\mathcal{I}} = 0.3$, what is $(a \vee b \rightarrow a)^{\mathcal{I}}$?

A natural choice:

- $(\neg a)^{\mathcal{I}} = 1 - a^{\mathcal{I}}$
- $(a \wedge b)^{\mathcal{I}} = \max(a^{\mathcal{I}}, b^{\mathcal{I}})$
- $(a \vee b)^{\mathcal{I}} = \min(a^{\mathcal{I}}, b^{\mathcal{I}})$

This choice is natural, but is too limited.

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What is deductive inference?

Inference is the process of deriving new knowledge out of a knowledge base.

Examples:

- In propositional logic: $\{a, a \rightarrow b\} \vdash b$
- In first order logic: $\{\forall x\ P(x) \rightarrow Q(x), P(a)\} \vdash Q(a)$

These examples of inference are called **deductive**: when the premises (here the knowledge base) are correct, then the conclusions are correct.

By contraposition, this also means that an incorrect conclusion means that at least one of the premises is incorrect.

Hypothetic inferences

Hypothetic inferences are forms of inference which do not guarantee that the conclusions are correct even when the premises are correct.

Why would we want that?

Hypothetic inferences

Hypothetic inferences are forms of inference which do not guarantee that the conclusions are correct even when the premises are correct.

Why would we want that?

Because...

- In many situations we do not have enough knowledge to make a decision, and yet we must make one.
- We may have enough knowledge, but not the deductive power necessary to extract the conclusion (when even possible, hello Gödel...)

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- Inductive generalization
- Analogical inference and case-based reasoning
- Abductive inference

Inductive generalization

Inductive generalization is a form of hypothetic reasoning which aims at generalizing the premises.

Example: Knowing that `rains(Hamburg, June_1)`, `rains(Hamburg, June_2)`, ..., `rains(Hamburg, June_12)`, one *infers* that $\forall d$ `date(d)` $\wedge$ `rains(Hamburg,` d`)`.

Note that this is not a valid conclusion from a deductive point of view: to have a valid deductive inference, one would need to have a knowledge over *all* possible days.

Deducive vs inductive inference

Deductive reasoning:
In Scotland, at least one sheep is at least half black.

Inductive reasoning:
In Scotland, all sheeps are black.

Overgeneralization

Inductive inference might conclude here that birds don't fly.

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- Abductive inference

Analogical reasoning

An extreme case of inductive generalization is when we have access to one premise only.

Analogical reasoning treats this case by assuming that if two entities are similar on several aspects, they should be similar on other aspects.

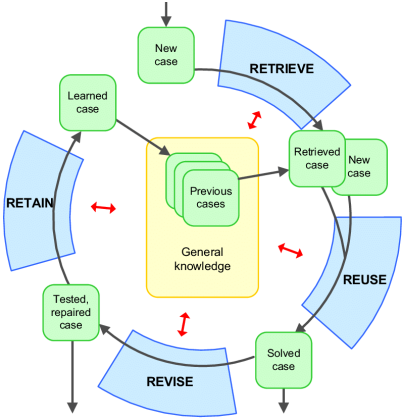
Example: If:

- `likes(Alice, The_Beatles)` $\wedge$ `likes(Alice, David_Bowie)`
- `likes(Alice, ABBA)`
- `likes(Bob, The_Beatles)` $\wedge$ `likes(Bob, David_Bowie)`

then by analogy we can infer that `likes(Bob, ABBA)`

Case-based reasoning

Case-based reasoning is a form of reasoning which generalizes analogical reasoning to a larger knowledge base.



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Inversing the cause and the consequences

In a situation where *cause* $\models$ *consequence* and where the consequence is observed, **abductive reasoning** aims at inferring the cause.

Example: Medical diagnosis is a typical example of abductive reasoning. Knowing that

$$covid_19 \rightarrow fever \wedge cough \wedge headache \wedge taste_loss$$

and observing the symptoms $fever \wedge cough \wedge headache \wedge taste_loss$ in a patient, abductive reasoning will infer that the patient has covid:

$$fever \wedge cough \wedge headache \wedge taste_loss \vdash_{abduction} covid_19$$

To know more...

Higher-order logic:

- Väänänen, J. (2019). Second-order and Higher-order Logic.

Modal logic:

- Garson, J. (2000). Modal logic.

Fuzzy logic

- Zadeh, L. A. (1988). Fuzzy logic.

Reasoning:

- Gentner, D. and Maravilla, F. (2017). Analogical reasoning.
- Gärdenfors, P. (1990). Belief revision and relevance.